Book 1A Chapter 5 Numerical Estimation

Warm-up Exercise

1. Round off each of the following numbers to 1 decimal place.

(a) 2.875

(b) 4.986

2. Round off each of the following numbers to 2 decimal places.

(a) 3.566 4

(b) 7208.921

3. Round off each of the following numbers to the nearest integer.

(a) 63.498

(b) 1572.6

4. Round off each of the following numbers to the nearest hundred.

(a) 5463

(b) 82 539

5. Round off each of the following numbers to the nearest ten.

(a) 16 505

(b) 98.347

6. Round off each of the following numbers to the nearest million.

(a) 2 499 999

(b) 713 560 489

5.1 Introduction to Numerical Estimation

Level 1

Demonstration 1

Estimate $4.5 + 8.2 + 6.7$ by rounding off each number to the nearest integer.

Solution

$4.5 + 8.2 + 6.7$

$\approx 5 + 8 + 7$

$= \underline{\underline{20}}$

1. Estimate $2.8 + 9.3 - 4.6$ by rounding off each number to the nearest integer.

Demonstration 2

Estimate $20.2 \times 11.9 + 43.7$ by rounding down each number to the nearest integer.

Solution

$20.2 \times 11.9 + 43.7$

$\approx 20 \times 11 + 43$

$= \underline{\underline{263}}$

2. Estimate $13.5 + 6.4 \times 8.7$ by rounding down each number to the nearest integer.

Demonstration 3

Estimate $15.6 - 44.1 \div 8.8$ by rounding up each number to the nearest integer.

Solution

$15.6 - 44.1 \div 8.8$

$\approx 16 - 45 \div 9$

$= \underline{\underline{11}}$

3. Estimate $27.5 \div 6.2 \times 8.01$ by rounding up each number to the nearest integer.

Level 2

4. Estimate the value of each of the following expressions by rounding off each number to the nearest integer.

   (a) $6.2 + 7.6 + 4.5$
   (b) $12.7 + 13.1 + 19.9$
   (c) $2.2 + 17.8 - 6.9$
   (d) $1.65 - 8.34 + 15.28$
   (e) $14.84 - 2.97 - 8.35$
   (f) $19.23 + 7.45 - 25.76$

5. Estimate the value of each of the following expressions by rounding off each number to the nearest integer.

(a) 3.8×5.13
(b) $16.4 \times 3.95 - 45.8$
(c) $2.69 + 6.12 \times 1.04$
(d) $19.5 \div 5.37 - 1.76$
(e) $14.78 \times 2.05 \div 5.93$
(f) $8.76 \div 2.55 \times 11.64$

6. Estimate the value of each of the following expressions by rounding down each number to the nearest integer.

(a) $6.6 + 9.3 + 2.5$
(b) $8.7 - 6.4 + 3.3$
(c) $12.5 + 7.4 - 8.16$
(d) $26.8 + 9.34 - 7.2$
(e) $4.56 + 2.85 - 10.39$
(f) $17.43 - 12.6 - 35.9$

7. Estimate the value of each of the following expressions by rounding down each number to the nearest integer.

(a) $18.24 \div 6.34$
(b) $7.32 \times 3.19 - 15.48$
(c) $36.6 \div 12.3 + 4.97$
(d) $39.5 - 42.4 \div 2.70$
(e) $12.54 + 7.08 \times 6.92$
(f) $5.32 \times 6.11 \div 2.18$

8. Estimate the value of each of the following expressions by rounding up each number to the nearest integer.

(a) $18.67 + 4.34 + 16.23$
(b) $13.75 + 24.53 - 19.07$
(c) $9.43 - 6.84 - 5.27$
(d) $52.43 - 67.5 + 46.8$
(e) $39.2 - (17.3 + 29.03)$
(f) $7.69 - (8.32 - 15.47)$

9. Estimate the value of each of the following expressions by rounding up each number to the nearest integer.

(a) $6.35 \times 1.78 - 8.24$
(b) $15.1 \div 3.89 + 12.92$
(c) $47.29 \div (22.13 - 14.56)$
(d) $3.35 \times (6.4 + 8.57)$
(e) $89.65 \div (62.45 \div 6.09)$
(f) $79.4 \div (3.75 \times 4.02)$

10. Estimate the value of $68.71 + 19.20 - 24.73$ by rounding off each number to

(a) the nearest ten,
(b) the nearest integer.

11. Estimate the value of $75.42 - (38.34 - 90.28)$ by rounding down each number to

(a) the nearest ten,
(b) the nearest integer.

12. Estimate the value of $19.34 \times 12.87 - 15.6$ by rounding up each number to

(a) the nearest ten,

(b) the nearest integer.

13. Estimate the value of $13.69 - 22.83 + 14.47$ by

(a) rounding off each number to the nearest integer,

(b) rounding down each number to the nearest integer,

(c) rounding up each number to the nearest integer.

14. Estimate the value of $38.64 + 10.23 - 27.19$ by

(a) rounding off each number to the nearest ten,

(b) rounding off each number to the nearest integer,

(c) rounding off each number to 1 decimal place.

Judge whether the estimated value of each of the following expressions is reasonable without calculating the exact value of the expression. **(15 – 18)**

15. $432 + 159 + 186 \approx 570$

16. $7.52 - 3.48 + 16.05 \approx 18$

17. $96 \times 82 - 78 \times 150 \approx -4000$

18. $16.5 + 83.7 \div 5.8 \approx 60$

19. In a city, the population is 2 567 825 and the number of hospitals is 20. Estimate the average number of citizens each hospital serves by rounding off the population to

(a) the nearest million,

(b) the nearest hundred thousand.

20. A gardener records the heights of six plants. Their heights are 1.32 m, 1.09 m, 0.72 m, 1.28 m, 0.96 m and 1.84 m. Estimate the total height of the plants by rounding down the height of each plant to 1 decimal place.

21. The number of residents in eight public housing estates are 7548, 9560, 6823, 12 357, 8759, 16 720, 8302 and 10 254.

(a) Estimate the total number of residents in the eight estates by rounding up the number of residents in each estate to the nearest thousand.

(b) Hence, estimate the average number of residents in each estate.

22. In a city, the numbers of male and female citizens aged 70 or above are 233 600 and 308 972 respectively.

(a) Estimate the total number of citizens aged 70 or above by rounding off the numbers of citizens to the nearest ten thousand.

(b) There are 62 community centres in the city. The government estimates that each community centre serves 9000 citizens aged 70 or above on average. Do you think the official's estimate is reasonable? Explain your answer.

23. In a university, the numbers of male and female undergraduates are 4835 and 6920 respectively.

(a) Estimate the total number of undergraduates by rounding off each number to the nearest thousand.

(b) It is given that the numbers of male and female undergraduates majoring in business and management are 270 and 532 respectively.

(i) Estimate the total number of undergraduates majoring in business and management by rounding off each number to the nearest hundred.

(ii) The spokesman of the university claims that about one-tenth of the undergraduates major in business and management. Do you agree? Explain your answer.

24. The price and the quantity of each dish of dim sum Ron orders in a restaurant are shown below.

Dim Sum	Quantity	Dim Sum	Quantity
Shrimp dumplings **Price: $24.6**	2	*Phoenix claws* **Price: $14.8**	1
Shaomai **Price: $18.3**	3	*Shrimp rice noodle rolls* **Price: $24.6**	1
Shanghai steamed buns **Price: $18.3**	1	*Cha siu buns* **Price: $14.8**	2
Steamed meat balls **Price: $14.8**	1	*Turnip cake* **Price: $14.8**	1

Estimate the total price of all the dishes of dim sum by rounding off each price

(a) to the nearest dollar,

(b) to the nearest ten dollars.

25. Ronald wants to buy six items priced at \$17.5, \$60.2, \$35.4, \$12.3, \$29.5 and \$8.4.

(a) Estimate the total price of the six items by

(i) rounding off the price of each item to the nearest dollar,

(ii) rounding down the price of each item to the nearest dollar,

(iii) rounding up the price of each item to the nearest dollar.

(b) Ronald wants to know if he can afford the six items. Which estimation should he use? Explain your answer.

26. Eight students want to buy some stuff to decorate their classroom. They have \$32.4, \$18.3, \$12.6, \$8.8, \$9.5, \$20, \$11.7 and \$4.5.

(a) Estimate the total amount they have by

(i) rounding off the amount each of them has to the nearest dollar,

(ii) rounding down the amount each of them has to the nearest dollar,

(iii) rounding up the amount each of them has to the nearest dollar.

(b) Using the result of **(a)**, explain if they have enough money to buy the stuff which costs \$110 in total.

Level 3

27. Jennifer measures the interior angles of a triangle. She estimates the sum of the interior angles by rounding off each angle to the nearest degree.

Open-ended **(a)** Suggest a set of possible angles of the triangle if her estimation is less than 180°.

(b) If she rounds up each angle to the nearest degree, can her estimation of the sum of the angles exceed 190°? Explain your answer.

Multiple Choice Questions

28. Estimate $5 \times 2.4 + 6 \times 2.5$ by rounding off each number to the nearest integer.

A. 22

B. 25

C. 27

D. 28

29. The prices of five gold rings are \$3450, \$8320, \$6280, \$2540 and \$5660. Estimate the average price of the rings by rounding down their prices to the nearest hundred dollars.

A. \$5200

B. \$5260

C. \$5300

D. \$5340

30. The prices of a packet of potato chips and a cup noodle are $13.9 and $8.4 respectively. Jason has $98. In order to make sure that he has enough money to buy 5 packets of potato chips and 3 cup noodles, which of the following is/are a suitable step in the estimation?

I. Round off each price to the nearest integer first.

II. Round up each price to the nearest integer first.

III. Round down each price to the nearest integer first.

A. I only

B. II only

C. III only

D. I and II only

5.2 More about Numerical Estimation

Level 1

Demonstration 1

Estimate the value of each of the following expressions by using compatible numbers.

(a) $22.943 \div 8.105$

(b) 0.664×117

Solution

(a) $22.943 \div 8.105 \approx 24 \div 8 = \underline{\underline{3}}$

(b) $0.664 \times 117 \approx \frac{2}{3} \times 120 = \underline{\underline{80}}$

1. Estimate the value of each of the following expressions by using compatible numbers.

(a) $75 \div 26$

(b) 45.8×0.393

Demonstration 2

Estimate the value of $5.93 + 6.04 + 5.88 + 5.85$ by using a clustered value.

Solution

Since the numbers are close to 6, we choose 6 as the clustered value.

$$\begin{aligned} & 5.93 + 6.04 + 5.88 + 5.85 \\ \approx\ & 6 \times 4 \\ =\ & \underline{\underline{24}} \end{aligned}$$

2. Estimate the value of $2.48 + 2.52 + 2.43 + 2.59$ by using a clustered value.

Demonstration 3

Estimate the value of each of the following expressions by rearranging the order of operations.

(a) $840 + 530 - 643 - 227$

(b) $1230 \times 489 \div 607$

Solution

(a)
$$\begin{aligned} & 840 + 530 - 643 - 227 \\ =\ & 840 - 643 + 530 - 227 \\ \approx\ & 840 - 640 + 530 - 230 \\ =\ & 200 + 300 \\ =\ & \underline{\underline{500}} \end{aligned}$$

(b)
$$\begin{aligned} & 1230 \times 489 \div 607 \\ =\ & \frac{1230 \times 489}{607} \\ \approx\ & \frac{1200 \times 500}{600} \\ =\ & 2 \times 500 \\ =\ & \underline{\underline{1000}} \end{aligned}$$

3. Estimate the value of each of the following expressions by rearranging the order of operations.

(a) $22.3 + 34.7 + 68.1 + 15.8$

(b) $3.89 \div 16.14 \times 19.27$

Level 2

4. Estimate the value of each of the following expressions by using compatible numbers.

(a) $\frac{55}{8}$ (b) $\frac{100}{19}$

(c) $\frac{63}{16}$ (d) $\frac{197}{25}$

(e) $\frac{97}{33}$ (f) $\frac{237}{58}$

5. Estimate the value of each of the following expressions by using compatible numbers.

(a) 23.57×4 (b) $19.13 \div 9$

(c) $35.05 \div 12.29$ (d) 68.43×50.75

(e) $98.3 \div 0.512$ (f) 0.196×25.08

6. Estimate the value of each of the following expressions by using compatible numbers.

(a) 49.22×62.03 (b) $23.87 \div 39.15$

(c) 0.667×718 (d) 0.334×83.55

(e) $19.83 \div 0.257 \times 4.96$ (f) $16.42 \times 25.08 \div 0.81$

Estimate the value of each of the following expressions by using a clustered value. (7 – 10)

7. $9.6 + 9.5 + 10.3 + 10.6 + 9.4 + 10.2$

8. $14.8 + 14.7 + 14.5 + 15.2 + 15.2 + 15.1 + 15.2$

9. $64.9 + 65.2 + 65.3 + 66.1 + 64.4 + 64.1$

10. $830 + 792 + 803 + 794 + 811 + 788 + 764 + 810$

11. Estimate the value of each of the following expressions by rearranging the order of operations.

(a) $43 - 73 + 66 - 19$ (b) $286 - 462 - 135 + 710$

12. Estimate the value of each of the following expressions by rearranging the order of operations.

(a) $619 - 752 + 333 - 246 - 108 + 855$ (b) $993 + 542 - 302 + 667 - 158 - 744$

13. Estimate the value of each of the following expressions by rearranging the order of operations.

(a) $240 \times 159 \div 61$ (b) $0.198 \div 307 \times 1825$

14. Judge whether the estimation $265 + 253 + 259 + 241 + 254 + 248 \approx 3000$ is reasonable without calculating the exact value of the expression.

15. Judge whether the estimation $74.6 + 75.5 + 75.2 + 73.3 + 75.8 + 75.9 + 76.3 + 76.3 \approx 600$ is reasonable without calculating the exact value of the expression.

16. Judge whether the estimation $48.5 + 8.3 \times 6 + 49.3 + 52.3 + 51.3 \approx 180$ is reasonable without calculating the exact value of the expression.

17. Judge whether the estimation $9073 + 18\,924 + 9162 + 8915 + 19\,047 + 8821 \approx 200\,000$ is reasonable without calculating the exact value of the expression.

18. Judge whether the estimation $24.18 \times 4.23 - 51.34 \approx 50$ is reasonable without calculating the exact value of the expression.

19. Judge whether the estimation $3.62 \times 1.93 + 4.42 \div 1.58 \approx 25$ is reasonable without calculating the exact value of the expression.

20. Judge whether the estimation $3520 \times 79 \div 48 \approx 5600$ is reasonable without calculating the exact value of the expression.

21. The weights of five parcels are 410 g, 420 g, 415 g, 427 g and 422 g. Estimate the total weight of the parcels by using a clustered value.

22. On a flag day, the amounts collected by six volunteers are \$80, \$78.3, \$81.9, \$69.3, \$79 and \$60.8. Estimate the total amount collected by using a clustered value.

23. Karen bought a series of seven novels of 383, 405, 395, 412, 380, 418 and 490 pages. She has read 958 pages. Estimate the total number of pages she has not yet read by using a clustered value.

24. A wholesaler buys 5830 eggs. 684 eggs are rotten and are dumped. The wholesaler then buys 3157 good eggs and sells 6279 eggs. Estimate the number of eggs left by rearranging the order of operations.

25. Emily buys an air-conditioner priced at \$5970 and she splits the payment into 24 installments. Estimate the amount of each installment by using compatible numbers.

26. Estimate the capacity of a swimming pool whose dimensions are 25 m × 39 m × 1.5 m by using compatible numbers.

27. A restaurant sells 792 bowls of noodles each week. Estimate the total number of bowls of noodles sold in 23 days by using compatible numbers.

28. The prices of each tray of snacks and spaghetti are \$49 and \$91.8 respectively. 19 trays of snacks and 5 trays of spaghetti are bought and their total price is shared equally among 11 people. Estimate how much each person should pay by using compatible numbers.

29. Suppose 12.478 yen can exchange for 1 HKD and 1.2131 HKD can exchange for 1 RMB.

(a) Estimate how much yen for which 3970 HKD can exchange.

(b) Estimate how much RMB for which 6025 HKD can exchange.

30. Angus buys six electrical and electronic appliances whose prices are shown below.

Item	Price
A TV set	\$3409
A washing machine	\$3786
A rice cooker	\$1227
A notebook computer	\$4583
A refrigerator	\$4650
An oven	\$928

(a) Estimate the total price of the electrical and electronic appliances.

(b) If he spends more than \$15 000, he will receive a digital camera for free. Estimate whether he can receive the digital camera for free.

31. Susan's height is 1.6 m. Her BMI (b) is given by the formula $b = \frac{w}{h^2}$, where w kg is her weight and h m is her height.

(a) If her weight is 60.4 kg, estimate her BMI by using compatible numbers.

(b) If her BMI is 19.3, estimate her weight by using compatible numbers.

Level 3

32. The table below records the prices of seven cars, which are arranged from left to right in ascending order. However, some of the prices of cars are stained.

Car	*A*	*B*	*C*	*D*	*E*	*F*	*G*
Price	\$245 900	\$417 600	\$4[illegible]0	\$4[illegible]200	\$4[illegible]00	\$433 700	\$625 400

Describe how you can estimate the total price of the seven cars.

Multiple Choice Questions

33. Eric is practising 100 m running. He records his times as follows:

10.2 s, 9.8 s, 9.5 s, 10.4 s, 9.9 s, 10.9 s, 10.5 s, 9.2 s

Estimate the average time by using a clustered value.

A. 9 s
B. 10 s
C. 11 s
D. 12 s

34. Suppose that 1200 yen can exchange for 9 Euro. Estimate how much Euro for which 23 680 yen can exchange by using compatible numbers.

A. 50 Euro
B. 180 Euro
C. 300 Euro
D. 400 Euro

35. Which of the following estimations is/are unreasonable?

I. $9.78 \times 5.13 \approx 168$
II. The height of International Financial Centre (412 m) is about 1.2 times the height of Tokyo tower (333 m).
III. The Earth takes about 10 000 hours to make one complete revolution around the Sun.

A. I only
B. I and II only
C. I and III only
D. II and III only